

Girsanov-Corrected Sequential Monte Carlo for Guided Diffusion Models

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1 Setup

1.1 Setting

We consider the Bayesian inverse problem of drawing samples from the posterior

$$p(x_0 \mid y) \propto p(y \mid x_0) p(x_0), \quad (1)$$

where $x_0 \in \mathbb{R}^d$ and y denotes observations. The prior $p(x_0)$ is given by a diffusion model. The likelihood $p(y \mid x_0)$ is assumed known.

1.2 Reverse diffusion and Girsanov correction

A diffusion model provides a time-reversed process that, when integrated from a pure-noise initialisation to zero noise, generates samples $x_0 \sim p(x_0)$. In continuous time this process is an Itô SDE

$$dx_\tau = a(x_\tau, \tau) d\tau + g(\tau) dW_\tau^\mathbb{P}, \quad (2)$$

where τ is the reverse-time parameter, $a(\cdot, \cdot)$ is the drift, and $g(\tau) \in \mathbb{R}^{d \times d}$ is the diffusion matrix. $W^\mathbb{P}$ is a d -dimensional Brownian motion under the measure $\mathbb{P}$. The path law of this process is denoted by $\mathbb{P}$.

To condition on y we require the intermediate likelihood $p(y \mid x_\tau) = \int p(y \mid x_0) p(x_0 \mid x_\tau) dx_0$, which is intractable; in the PDE setting a surrogate is used instead (Section 3). We assume $p(y \mid x_\tau)$ is positive and differentiable in x_τ . Define the guidance drift

$$b(x_\tau, \tau) := \nabla_{x_\tau} \log p(y \mid x_\tau). \quad (3)$$

The guided reverse process augments the drift by $\Sigma(\tau) b(x_\tau, \tau)$ where $\Sigma(\tau) := g(\tau)g(\tau)^\top$, retaining the same diffusion matrix g :

$$dx_\tau = [a(x_\tau, \tau) + \Sigma(\tau) b(x_\tau, \tau)] d\tau + g(\tau) dW_\tau^\mathbb{Q}, \quad (4)$$

with path law $\mathbb{Q}$. The two processes share the same g ; they differ *only* in their drift.

Girsanov's theorem (Øksendal, 2003, Theorem 8.6.6) states that $\mathbb{P}$ and $\mathbb{Q}$ are equivalent. Because the drift difference $\Sigma(\tau) b(x_\tau, \tau)$ lies in the column space of $g(\tau)$, the Radon-Nikodym derivative is

$$\log \frac{d\mathbb{P}}{d\mathbb{Q}} = - \int b^\top (g dW^\mathbb{Q}) - \frac{1}{2} \int b^\top \Sigma b d\tau. \quad (5)$$

We assume the usual regularity conditions for Girsanov's theorem (progressive measurability of $g^\top b$ and a Novikov-type integrability condition).

1.3 Discretization

The particle filter operates in discrete time, so we partition the reverse trajectory into K steps. The reverse-time parameter τ is oriented so that integration over the step runs from τ_k to τ_{k-1} (a positive increment). Over step k we denote the accumulated diffusion matrix and the Brownian increment by

$$\Sigma_k := \int_{\tau_k}^{\tau_{k-1}} \Sigma(\tau) d\tau, \quad z_k := \int_{\tau_k}^{\tau_{k-1}} g(\tau) dW_\tau^{\mathbb{Q}} \sim \mathcal{N}(0, \Sigma_k), \quad (6)$$

where z_k is the noise drawn during the proposal step.

The per-step contribution to the Radon-Nikodym derivative is the restriction of the integrals in (5) to $[\tau_k, \tau_{k-1}]$:

$$- \int_{\tau_k}^{\tau_{k-1}} b^\top (g dW^{\mathbb{Q}}) - \frac{1}{2} \int_{\tau_k}^{\tau_{k-1}} b^\top \Sigma b d\tau. \quad (7)$$

The function $b(\tau)$ is known only at discrete points along the trajectory. The standard Itô discretisation of (7) evaluates the integrand at the left endpoint x_k , which is adapted to the Brownian filtration:

$$C_k := -b_k^\top z_k - \frac{1}{2} b_k^\top \Sigma_k b_k, \quad b_k := b(x_k, \tau_k). \quad (8)$$

More accurate Girsanov discretisations are catalogued in Appendix B. The left-endpoint rule is the simplest Itô-consistent discretisation and coincides exactly with the Gaussian kernel ratio when both $p(\cdot | x_k)$ and $q(\cdot | x_k)$ are Euler-Maruyama kernels (Appendix A).

1.4 Sequential Monte Carlo

The target measure over complete paths is $p(y | x_0) \mathbb{P}$. We use the guided path law $\mathbb{Q}$ as the proposal, so the path-level importance weight is $p(y | x_0) d\mathbb{P}/d\mathbb{Q}$. With the per-step Girsanov corrections C_k this factorises as

$$w(x_{0:K}) \propto p(y | x_0) \prod_{k=K}^1 \exp(C_k). \quad (9)$$

To obtain a sequential update, we factor $p(y | x_0)$ by telescoping over the schedule:

$$p(y | x_0) = p(y | x_K) \prod_{k=K}^1 \frac{p(y | x_{k-1})}{p(y | x_k)}. \quad (10)$$

(Products are taken over $k = K, K-1, \dots, 1$.) Substituting this into (9) gives

$$w \propto p(y | x_K) \prod_{k=K}^1 \frac{p(y | x_{k-1})}{p(y | x_k)} \exp(C_k). \quad (11)$$

Define the initial weight $w_K \propto p(y | x_K)$. Then at each step the incremental log weight is

$$\log w_{k-1} = \log w_k + [\log p(y | x_{k-1}) - \log p(y | x_k)] + C_k. \quad (12)$$

Resampling is triggered whenever the effective sample size drops below a threshold.

It is also useful to introduce two tunable parameters, allowing a continuous interpolation between weighting strategies:

$$\log w_{k-1} = \log w_k + \rho \left[\log p(y \mid x_{k-1}) - \log p(y \mid x_k) \right] + \lambda C_k, \quad (13)$$

where $\lambda \in \{0, 1\}$ switches the Girsanov correction on (1) or off (0), and $\rho > 0$ tempers the likelihood contribution. When $\rho \neq 1$, the initial weight is $w_K \propto p(y \mid x_K)^\rho$ to preserve the telescoping identity. The choice $\lambda = 1$, $\rho = 1$ yields the posterior $p(x_0 \mid y) \propto p(y \mid x_0) p(x_0)$, up to discretisation and Girsanov-quadrature error. The choice $\lambda = 0$ drops the path-measure correction and recovers the pseudo-bootstrap scheme of Millard et al. (2026), which is generally biased relative to the posterior target.

1.5 Proposals

The particle filter requires discretising two continuous-time objects: the particle dynamics (turning the guided SDE into a transition kernel q) and the per-step weight correction.

For any proposal $q(x_{k-1} \mid x_k)$, the ideal incremental weight is the kernel ratio $\log p(x_{k-1} \mid x_k)/q(x_{k-1} \mid x_k)$. Neither kernel is available in closed form for a general integrator. However, as $K \rightarrow \infty$ both converge to their continuous-time limits, and Girsanov’s theorem relates their ratio to the path integral of (5). The left-endpoint discretisation C_k of (8) thus provides a consistent weight approximation for any integrator: both $\log p/q$ and C_k converge to the same limit as $K \rightarrow \infty$.

For the dynamics we evaluate two integrators:

- **Guided Euler-Maruyama (GEM).** A first-order integrator producing a Gaussian proposal kernel. For Euler-Maruyama the left-endpoint C_k coincides with the exact kernel ratio identically at every step (Appendix A). With $\lambda = 1$ the weight is therefore identical to the Twisted Diffusion Sampler (Wu et al., 2023). One score evaluation per step.
- **Stochastic Heun (Heun-SDE).** A Runge-Kutta integrator for additive-noise SDEs, second-order in the drift. The first stage predicts via Euler; the second stage corrects by averaging the drift at x_k and the predicted midpoint x_{pred} ; both stages are driven by the same noise z_k . The transition is non-Gaussian, so the exact kernel ratio is intractable; C_k provides an asymptotically valid weight. Two score evaluations per step.

For the Girsanov correction, the left-endpoint rule C_k is the baseline quadrature. More accurate discretisations are catalogued in Appendix B; they can be combined with any integrator.

The particle filter inherits two discretisation errors: trajectory error from the integrator and weight error from the Girsanov quadrature. With the left-endpoint rule the weight accuracy is first-order for both proposals, so while Heun-SDE yields a more accurate particle population, both integrators produce comparably reliable weights. Given b_k , evaluating C_k adds only $\mathcal{O}(d)$ operations per particle per step, so the computational overhead is negligible compared to the score evaluations.

1.6 Outline

The remainder of the note validates this weighting machinery. Section 2 and 3 describe two application settings with their respective notation. Appendices A and B collect the Euler kernel-ratio verification and the catalogue of alternative Girsanov discretisations.

2 Toy model: Gaussian mixture

We validate the weighting machinery on a one-dimensional Gaussian-mixture model whose every ingredient is available in closed form: the exact score, the exact intermediate likelihood, and the analytic posterior. Any deviation of the SMC output from the analytic posterior is therefore algorithm error, not model error. All quantities in this section are computed in closed form from the mixture; no learned model is involved.

2.1 Model

Write the prior as a two-component mixture

$$p(x_0) = \sum_{m=1}^2 c_m p_m(x_0), \quad p_m(x_0) = \mathcal{N}(x_0; \mu_m, \sigma_m^2), \quad (14)$$

with component weights c_m , means μ_m , and variances σ_m^2 . The observation model is $p(y | x_0) = \mathcal{N}(y; x_0, \gamma^2)$. By Bayes' rule,

$$p(x_0 | y) \propto p(y | x_0) p(x_0) = \sum_{m=1}^2 c_m p_m(x_0) p(y | x_0). \quad (15)$$

Each product $p_m(x_0) p(y | x_0)$ is proportional to a Gaussian in x_0 , so

$$p(x_0 | y) \propto \sum_{m=1}^2 \hat{c}_m p_m(x_0 | y), \quad (16)$$

where $p_m(x_0 | y) = \mathcal{N}(x_0; \hat{\mu}_m, \hat{\sigma}_m^2)$ with $\hat{\mu}_m = \mu_m + \frac{\sigma_m^2}{\sigma_m^2 + \gamma^2}(y - \mu_m)$, $\hat{\sigma}_m^2 = \frac{\sigma_m^2 \gamma^2}{\sigma_m^2 + \gamma^2}$, and $\hat{c}_m \propto c_m p_m(y)$ with $p_m(y) = \mathcal{N}(y; \mu_m, \sigma_m^2 + \gamma^2)$.

2.2 Schedule

The reverse trajectory uses the EDM power-law schedule (Karras et al., 2022) over K steps, indexed $k = K, \dots, 1$: $s_k = (s_K^{1/r} + \frac{k}{K}(s_0^{1/r} - s_K^{1/r}))^r$ (concrete values of s_K , s_0 , r are given in Section 2.7). From (6) the per-step accumulated diffusion is

$$\delta_k =: \Sigma_k = s_k^2 - s_{k-1}^2, \quad (17)$$

and in one dimension (8) gives $C_k = -b_k z_k - \frac{1}{2} \delta_k b_k^2$.

2.3 Noising and exact score

With variance-exploding additive noise, $p(x_k | x_0) = \mathcal{N}(x_k; x_0, s_k^2)$. The noised marginal is therefore $p(x_k) = \sum_{m=1}^2 c_m p_m(x_k)$, where $p_m(x_k) = \mathcal{N}(x_k; \mu_m, \sigma_m^2 + s_k^2)$. The score is

$$\nabla_{x_k} \log p(x_k) = \sum_{m=1}^2 \alpha_m(x_k) \frac{\mu_m - x_k}{\sigma_m^2 + s_k^2}, \quad \alpha_m(x_k) = \frac{c_m p_m(x_k)}{p(x_k)}. \quad (18)$$

2.4 Intermediate likelihood and guidance

Conditioned on component m and state x_k ,

$$p_m(y | x_k) = \mathcal{N}\left(y; \mu_m + \frac{\sigma_m^2}{\sigma_m^2 + s_k^2}(x_k - \mu_m), \gamma^2 + \frac{\sigma_m^2 s_k^2}{\sigma_m^2 + s_k^2}\right). \quad (19)$$

The intermediate likelihood is therefore

$$p(y | x_k) = \sum_{m=1}^2 \alpha_m(x_k) p_m(y | x_k). \quad (20)$$

Guidance follows (3); b_k is computed by differentiating (20). Because $\alpha_m(x_k)$ are genuinely nonlinear in x_k , the Hessian and divergence are non-constant, so the alternative Girsanov discretisations of Appendix B are meaningfully distinguishable (a single-component prior would collapse all of them).

2.5 Sequential Monte Carlo

Weighting follows the unified update (13). Particles are initialised from the exact noised marginal $p(x_K)$ (not white noise), so the initial weight is

$$w_K \propto p(y | x_K)^\rho, \quad (21)$$

the boundary term required by the telescoping identity once $\rho \neq 1$. Resampling is triggered whenever the effective sample size drops below half the particle count (Chopin and Papaspiliopoulos, 2020).

2.6 Metrics

In one dimension the Wasserstein-1 distance between the particle cloud and the analytic posterior admits a closed form,

$$W_1(\hat{p}, p(\cdot | y)) = \int_{\mathbb{R}} |\hat{F}(x) - F(x)| dx, \quad (22)$$

where $\hat{F}$ is the empirical CDF of the particles and F is the analytic posterior CDF obtained from (16). Alongside W_1 we report the errors of the weighted posterior mean and standard deviation, $|\hat{m} - m^*|$ and $|\hat{s} - s^*|$, the final effective sample size, and the number of resampling events. To attribute weight variance to its sources we also track the per-step decomposition

$$\text{Var}[\Delta \log w_k] = \rho^2 \text{Var}[\Delta \ell] + \lambda^2 \text{Var}[C_k] + 2\lambda\rho \text{Cov}[\Delta \ell, C_k], \quad (23)$$

with $\Delta \ell = \log p(y | x_{k-1}) - \log p(y | x_k)$.

2.7 Experiments

The model parameters are $c = (0.6, 0.4)$, $\mu = (-1.5, 2.0)$, $\sigma_m^2 = (0.7, 1.3)$, with $y = 0.0$ and $\gamma^2 = 1.0$; the resulting analytic posterior has mean -0.326 and standard deviation 1.061 . The schedule uses $s_K = 6$, $s_0 = 10^{-3}$, $r = 7$.

The implementation (`smc/toy_smc.py`) uses the GEM proposal with the left-endpoint correction (8). The following sweeps are run (see the code for details):

- **K -sweep.** $K \in \{100, 200, 500, 1000, 2000, 5000\}$, $N = 1024$, $(\lambda, \rho) = (1, 1)$.

- **λ sweep.** $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$, $K = 5000$, $N = 1024$, $\rho = 1$.
- **λ comparison K -sweep.** Same K values, comparing $\lambda = 1$ against $\lambda = 0$, $N = 1024$.

All runs initialise particles from the exact noised marginal $p(x_K)$ with the boundary weight (21). Reported metrics are W_1 distance (22), weighted-mean and weighted standard-deviation error, effective sample size, and resampling count.

Results

λ sweep ($K = 5000$, $N = 1024$). $|\text{dst}|$ drops monotonically from 0.29 ($\lambda = 0$) to 0.015 ($\lambda = 1$), a $19\times$ reduction, confirming that the Girsanov correction controls posterior variance recovery. W_1 improves from 0.26 to 0.03, and ESS remains above 84% of N for all λ (no resampling triggered).

K sweep ($\lambda = 1$, $N = 1024$). All metrics stay bounded as K grows. With GEM the left-endpoint correction (8) equals the exact kernel ratio at every step, so the weight is not a source of discretisation bias; W_1 oscillates around ~ 0.04 within Monte Carlo noise and no ESS collapse occurs.

λ comparison (same K values, $N = 1024$). At every K , $\lambda = 1$ improves W_1 by a factor of 5–8 and $|\text{dst}|$ by 10–100 $\times$ relative to $\lambda = 0$, for which $|\text{dst}| \approx 0.29$ independently of K . The Girsanov correction is essential for accurate posterior recovery.

Full figures are generated by the code (`smc/figs/`) and include comparison with the deterministic Heun ODE baseline, which collapses to a point estimate with no uncertainty quantification.

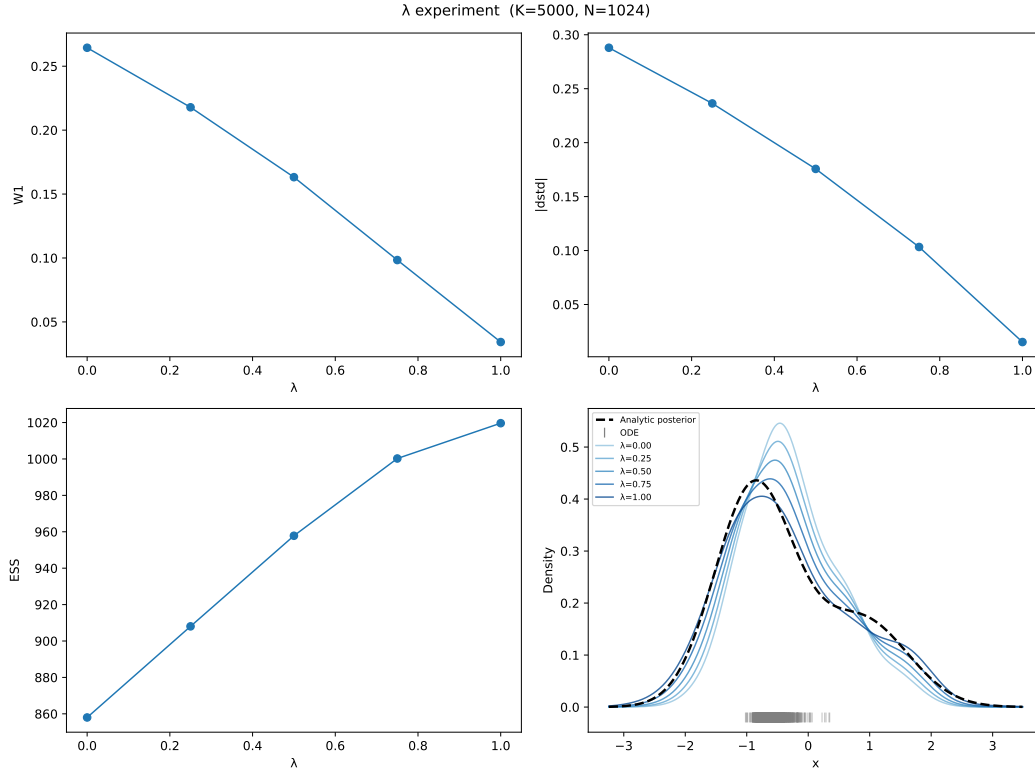


Figure 1: λ sweep ($K = 5000$, $N = 1024$).

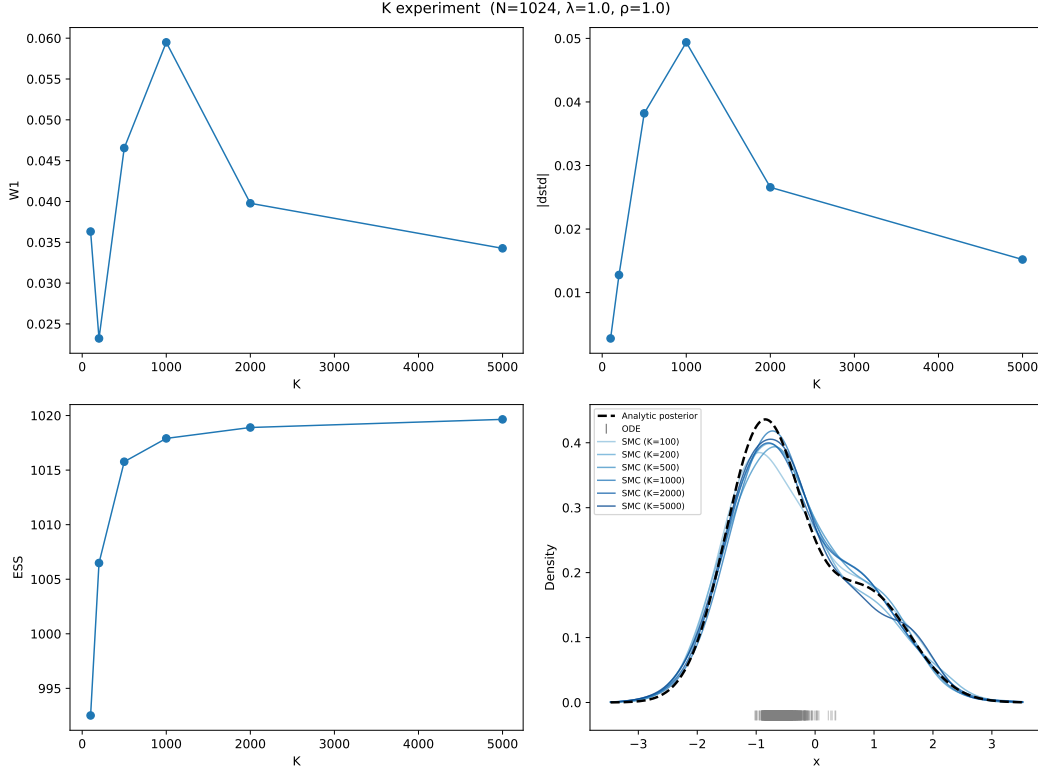


Figure 2: K sweep ($\lambda = 1$, $N = 1024$).

3 PDE-constrained reconstruction

The same machinery transfers to PDE-constrained reconstruction (Burgers and Darcy solvers), where the prior is a pretrained diffusion model over solution fields and the likelihood combines sparse sensor observations with a PDE residual. The exact intermediate likelihood of the toy, (20), is intractable in this setting; the guided samplers replace it with a surrogate formed from scaled observation and residual losses, whose gradient plays the role of the guidance drift (3). The weighting machinery of Section 2.7 transfers directly. Details of scaling, sensor masks, and the experimental design are deferred; the toy model above serves as the closed-form ground truth for the weighting machinery before any trained network is involved.

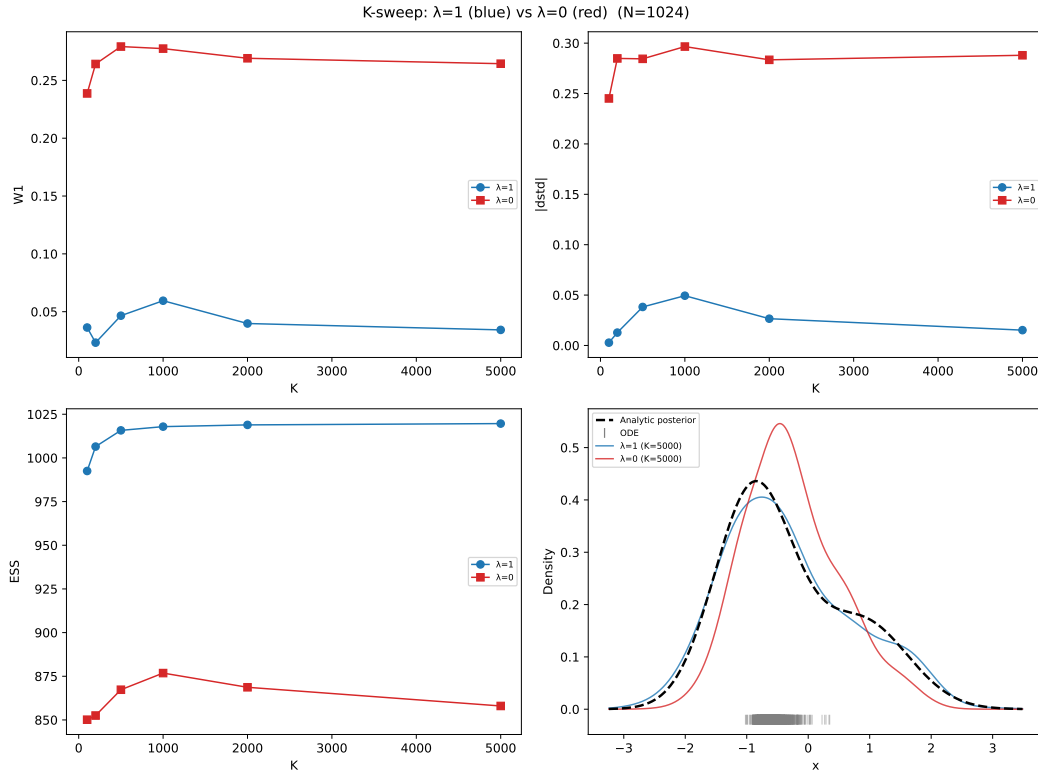


Figure 3: λ comparison K -sweep ($N = 1024$).

A Euler-Maruyama kernel ratio

For the Euler-Maruyama discretisation, both the unguided kernel $p(x_{k-1} \mid x_k)$ and the guided proposal $q(x_{k-1} \mid x_k)$ are Gaussian with common covariance $\delta_k I$, differing only in their means:

$$\begin{aligned} p(x_{k-1} \mid x_k) &= \mathcal{N}(x_{k-1}; \mu_p, \delta_k I), \\ \mu_p &:= x_k + \delta_k \nabla_{x_k} \log p(x_k), \\ q(x_{k-1} \mid x_k) &= \mathcal{N}(x_{k-1}; \mu_q, \delta_k I), \\ \mu_q &:= x_k + \delta_k (\nabla_{x_k} \log p(x_k) + b_k) = \mu_p + \delta_k b_k. \end{aligned}$$

Drawing x_{k-1} from q gives $x_{k-1} = \mu_q + \sqrt{\delta_k} z_k$ with $z_k \sim \mathcal{N}(0, I)$. The log density ratio is therefore

$$\begin{aligned} \log \frac{p(x_{k-1} \mid x_k)}{q(x_{k-1} \mid x_k)} &= -\frac{1}{2\delta_k} \left(\|x_{k-1} - \mu_p\|^2 - \|x_{k-1} - \mu_q\|^2 \right) \\ &= -\frac{1}{2\delta_k} \left(\|\sqrt{\delta_k} z_k + \delta_k b_k\|^2 - \|\sqrt{\delta_k} z_k\|^2 \right) \\ &= -\sqrt{\delta_k} b_k^\top z_k - \frac{1}{2} \delta_k \|b_k\|^2 = C_k. \end{aligned}$$

Thus, for the Euler-Maruyama discretisation, the Girsanov correction C_k equals the exact Gaussian kernel ratio identically at every step size δ_k .

B Alternative Girsanov discretisations

The left-endpoint rule (8) is the simplest Itô-consistent single-step discretisation of the Girsanov integral. This appendix catalogues several refinements that, though not used in the current experiments, remain practically testable. The Itô-formula discretisation (B.6) is implemented and validated in the companion toy model of `smc/toy_mixture.py` (see `smc/check_toy_mixture.py`).

B.1 Subdiscretised Girsanov correction

Split step k into m substeps with subincrements $\delta_{k,1}, \dots, \delta_{k,m}$ such that $\sum_{j=1}^m \delta_{k,j} = \delta_k$. Draw adapted Brownian subincrements $\Delta B_{k,j} = \sqrt{\delta_{k,j}} z_{k,j}$ and evaluate b before each subincrement is drawn. The per-step correction becomes

$$C_k^{\text{sub}} = -\sum_{j=1}^m \sqrt{\delta_{k,j}} b(x_{k,j})^\top z_{k,j} - \frac{1}{2} \sum_{j=1}^m \delta_{k,j} \|b(x_{k,j})\|^2. \quad (24)$$

This is Itô-adapted and converges to the continuous Girsanov correction as the submesh is refined. Cost: m additional evaluations of b per step.

B.2 Itô-Taylor correction

Let $H_k := \nabla_{x_k}^2 \log p(y \mid x_k)$ be the Hessian of $\log p$. A first-order Itô-Taylor expansion of the stochastic integral yields

$$C_k^{\text{IT}} = -\sqrt{\delta_k} b_k^\top z_k - \frac{1}{2} \delta_k \left[z_k^\top H_k z_k - \text{tr}(H_k) \right] - \frac{1}{2} \delta_k \|b_k\|^2. \quad (25)$$

Cost: Hessian-vector products and a trace estimate (in high dimensions, Hutchinson-type estimation is required for $\text{tr}(H_k)$). Following the substitution of Itô's formula to completion yields the Itô-formula discretisation of Appendix B.6, of which (25) is the first-order expansion.

B.3 Corrected midpoint and trapezoidal rules

Midpoint and trapezoidal stochastic sums approximate a Stratonovich integral. The conversion to an Itô integral for diffusion coefficient $g(\tau)I$ is

$$\int g b \cdot dW^{\mathbb{Q}} = \int g b \circ dW^{\mathbb{Q}} - \frac{1}{2} \int g^2 \nabla \cdot b \, d\tau,$$

where $\nabla \cdot b = \Delta_x \log p(y \mid x)$. A trapezoidal rule with the correction becomes

$$C_k^{\text{trap}} \approx -\frac{1}{2}(b_k + b_{k-1})^\top \Delta B_k + \frac{1}{2}\delta_k \nabla \cdot b_k - \frac{1}{2}\delta_k \|b_k\|^2.$$

A simpler hybrid keeps the left-endpoint rule for the stochastic term and applies a higher-order quadrature only to the ordinary time integral $-\frac{1}{2} \int g^2 \|b\|^2 d\tau$, e.g. using the predictor point:

$$C_k^{\text{hybrid}} = -\sqrt{\delta_k} b_k^\top z_k - \frac{1}{4}\delta_k (\|b_k\|^2 + \|b_{\text{pred}}\|^2).$$

This improves part of the correction while avoiding the Itô consistency issue in the stochastic term.

B.4 Approximate discrete kernel ratio

For a non-Gaussian proposal q , the transition is a deterministic function $x_{k-1} = F(x_k, z_k)$. If F is invertible in z_k ,

$$q(x_{k-1} \mid x_k) = \phi(z_k) \left| \det \frac{\partial x_{k-1}}{\partial z_k} \right|^{-1},$$

where ϕ is the standard Gaussian density. A corresponding evaluation for the unguided kernel $p(x_{k-1} \mid x_k)$ gives the discrete kernel ratio directly, targeting the finite-step proposal more accurately than a continuous-time Girsanov approximation. Cost: inverse maps and log-determinant computations, usually prohibitive in high dimensions; approximate log-determinants can be estimated with trace methods.

B.5 Heuristic predictor-point correction

Replacing b_k with b_{pred} in (8),

$$C_k^{\text{pred}} = -\sqrt{\delta_k} b_{\text{pred}}^\top z_k - \frac{1}{2}\delta_k \|b_{\text{pred}}\|^2,$$

is sometimes used as an empirical heuristic. Because x_{pred} depends on the same z_k , this is not an Itô-consistent discretisation without additional correction terms. Its performance can be tested experimentally but should not be claimed as an automatic higher-order Girsanov rule.

B.6 Itô-formula discretisation

Applying Itô's formula to $\ell_\tau(x) := \log p(y \mid x_{\sigma(\tau)} = x)$ under the guided process (4) eliminates the stochastic integral from (7) entirely. The per-step correction becomes

$$C_k^{\text{V}} := \int_{\tau_k}^{\tau_{k-1}} V_\tau(x_k) \, d\tau, \tag{26}$$

$$V_\tau(x) := \partial_\tau \ell_\tau(x) + a(x, \tau) \cdot \nabla \ell_\tau(x) + \frac{1}{2} \nabla \ell_\tau^\top \Sigma \nabla \ell_\tau + \frac{1}{2} \text{tr}(\Sigma \nabla^2 \ell_\tau(x)), \tag{27}$$

where the pre-step state x_k is frozen during the deterministic quadrature. The $\partial_\tau \ell_\tau$ term integrates exactly (telescoping): $\int_{\tau_k}^{\tau_{k-1}} \partial_\tau \ell_\tau d\tau = \ell_{\tau_{k-1}}(x_k) - \ell_{\tau_k}(x_k)$. In practice only the remaining part,

$$H_\tau(x) := a(x, \tau) \cdot \nabla \ell_\tau(x) + \frac{1}{2} \nabla \ell_\tau^\top \Sigma \nabla \ell_\tau + \frac{1}{2} \text{tr}(\Sigma \nabla^2 \ell_\tau(x)), \quad (28)$$

requires numerical quadrature, giving $C_k^V = \ell_{\tau_{k-1}}(x_k) - \ell_{\tau_k}(x_k) + \int_{\tau_k}^{\tau_{k-1}} H_\tau(x_k) d\tau$ (`smc/check_toy_mixture.py`).

In the variance-exploding parametrisation ($\Sigma = \bar{a}I$, $a(x, \tau) = \bar{a} \nabla \log p$, $\partial_\tau = -\partial_\sigma$, $\bar{a}(\tau) := 2\sigma(\tau)$) this reduces to $H_\tau = \bar{a} \nabla \log p \cdot \nabla \ell + \frac{1}{2} \bar{a} \|\nabla \ell\|^2 + \frac{1}{2} \bar{a} \Delta \ell$.

Unlike the left-endpoint rule (8) and its variants (24)–(25), (26) contains *no stochastic term* — the per-step weight is a deterministic function of x_k . The price is the Laplacian $\Delta \ell = \text{tr}(\nabla^2 \ell)$, which in high dimensions requires stochastic estimation (cf. Hutchinson-type trace estimators; `smc/hutchinson.py`). In one dimension $\Delta \ell$ reduces to an exact second derivative via double backward differentiation. The Itô-Taylor correction (25) is the first-order expansion of (26) that *retains* the stochastic term while adding a Hessian-based correction; C_k^V completes the substitution by eliminating it entirely.

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